without clear experimental evidence for something beyond the Standard Model, it's hard to know where to go next, or which of the various proposals are likely to be right. "There's all this beautiful stuff that's come out. But have we solved the problem of quantum gravity? I don't think so. And the problem is, how would we know if we'd solved it?"

Fortunately, no one is giving up hope. I spoke with dozens of cosmologists and particle physicists about where this whole thing is going (where by "whole thing," I mean both theoretical physics/cosmology and the actual universe), and while there was no agreement about the optimal approach, there were a few common themes. One was diversification: whatever big multinational experiments or observational programs we decide to invest in, it's important to diversify our approaches and come up with ideas that will give us new perspectives on these old problems (which goes for the theory side as well as the data-taking side). The other was the importance of continuing to get as much new data as we can, and to analyze it in every way possible.

Clifford V. Johnson, a theoretical physicist at the University of Southern California, works on string theory, black holes, extra dimensions of space, and the subtleties of entropy. He is about as deep into pure theory as anyone I know, and he is *very* excited about data right now. "My feeling is that we are maybe lacking a good sort of single idea, but we are not lacking in huge sources of data," he said. "And that reminds me of the immediate pre-quantum days, right?" In those days, theory was booming, with lots of half-formed ideas about the structure of atoms and nuclei, though none were all that compelling. "But we just then got all this wonderful data that began to eventually take shape. I don't see why that can't happen again. Looking at the history of science, that's how this works."

So let's talk about data. What we're looking at, and how, in both cosmology and particle physics. What it might tell us about both the physics of the universe today and how it's all going to come to an end in the future. And then we'll check back in with the theorists. Because some of the ideas they're talking about right now are absolutely wild.

TOUCHING THE VOID

If we want to learn anything about the far future of the cosmos, we'd better address the giant invisible ever-expanding killer elephant in the room: dark